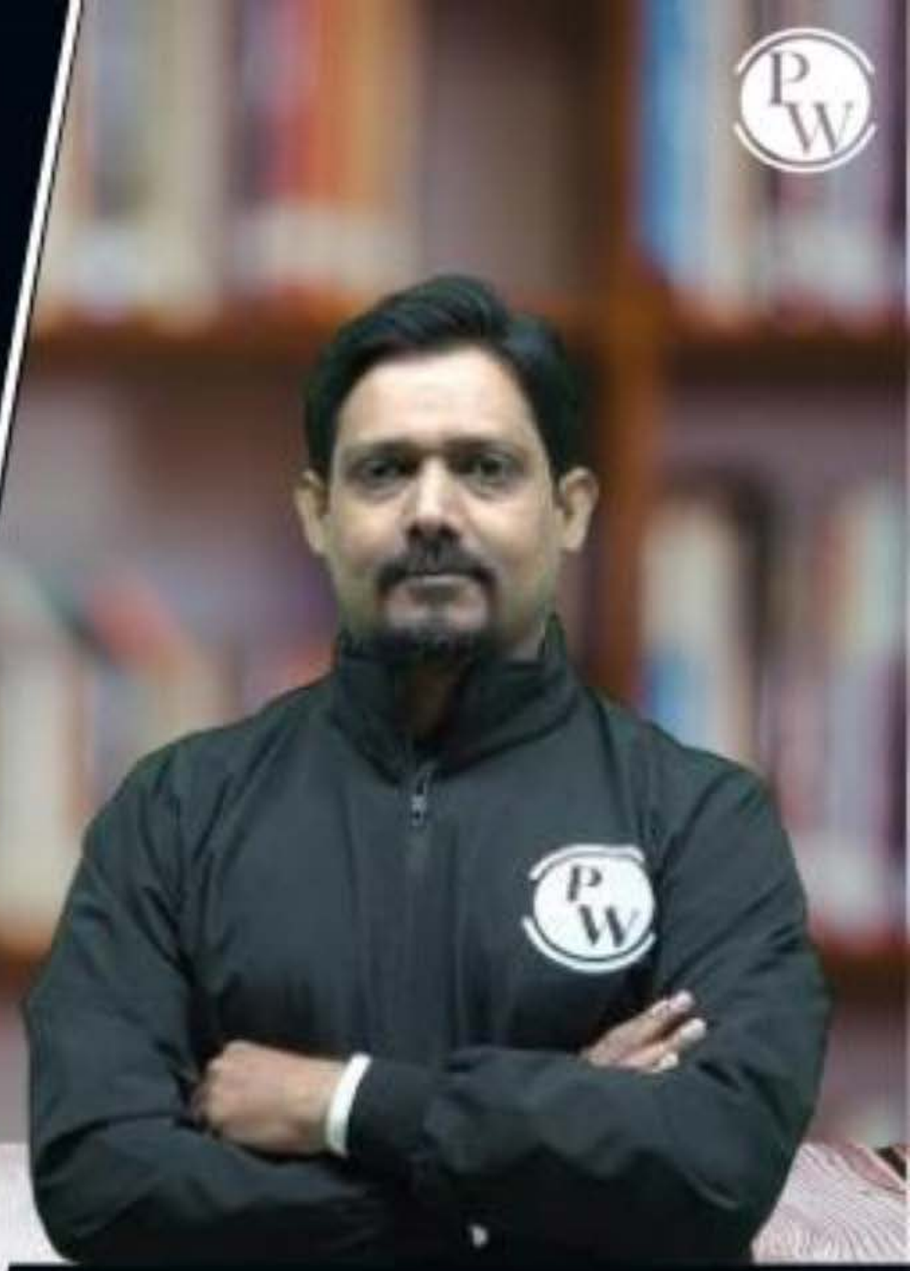
Computer Science & IT



Data Structure & Programming

Tree

Lecture No. 03



By- Abhishek Sir

Recap of Previous Lecture



Topic — Binary tree

Topic — Binay Tree height

Topic — Counting of unlabelled

Topic

Topic

Slide

Topics to be Covered



- Topic — Binay Tree Theorem
- Topic — Binay Tree representation
- Topic — Traversal
- Topic
- Topic

Slide



Topic : Question

3-ary

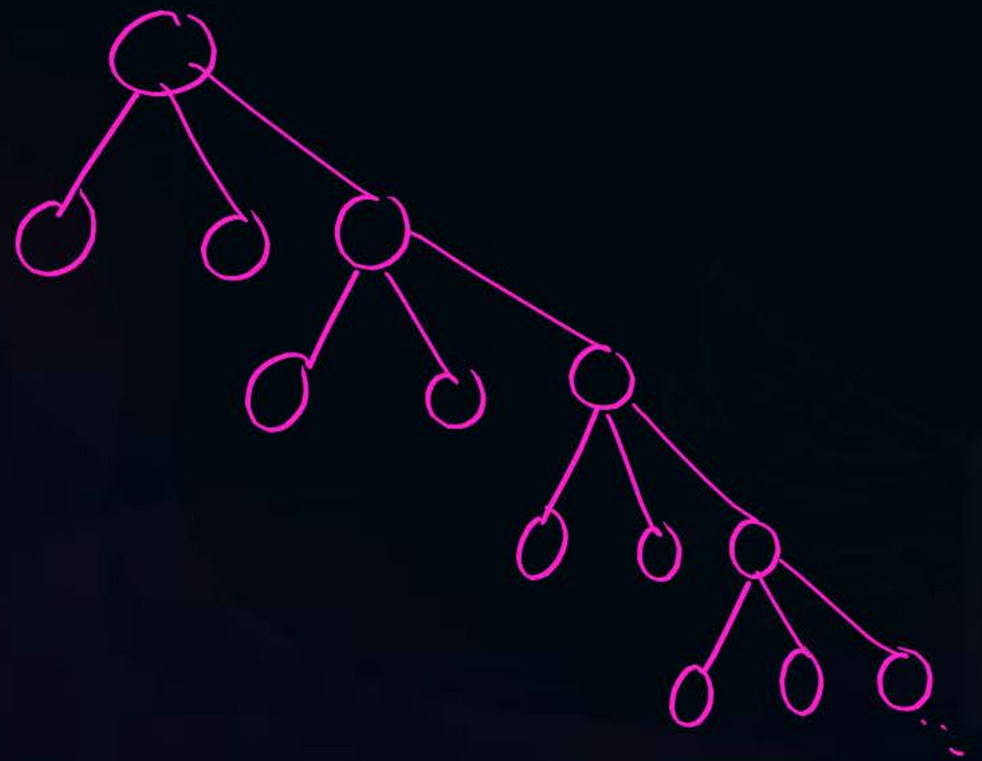
#Q Let T be complete ternary tree of height 6 where each node is having 0 or 3 children. The maximum and minimum number of nodes possible in T is (The root node is at height 0)?

(A) 13 and 3280

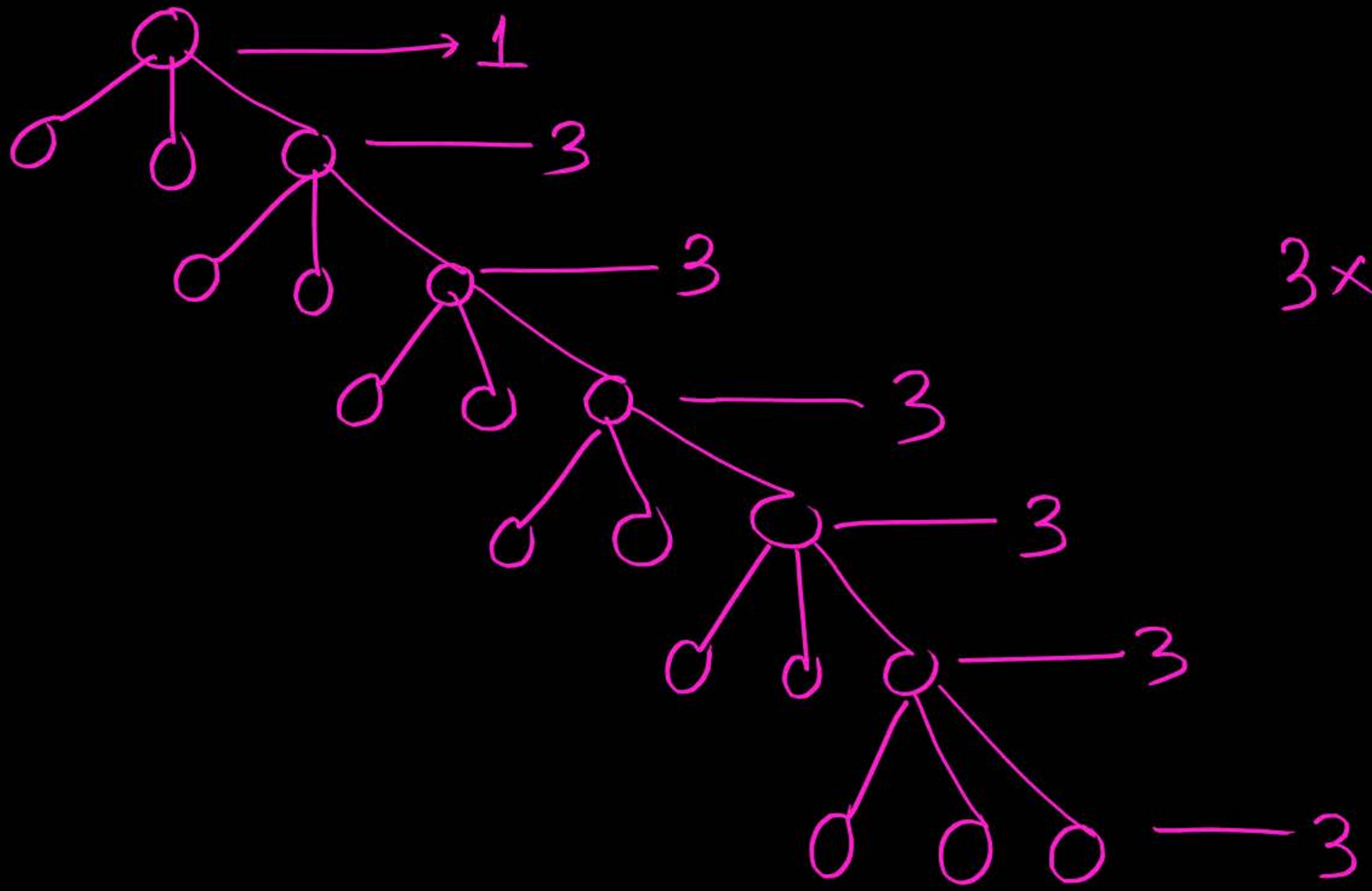
(B) 3280 and 13

(C) 1093 and 19

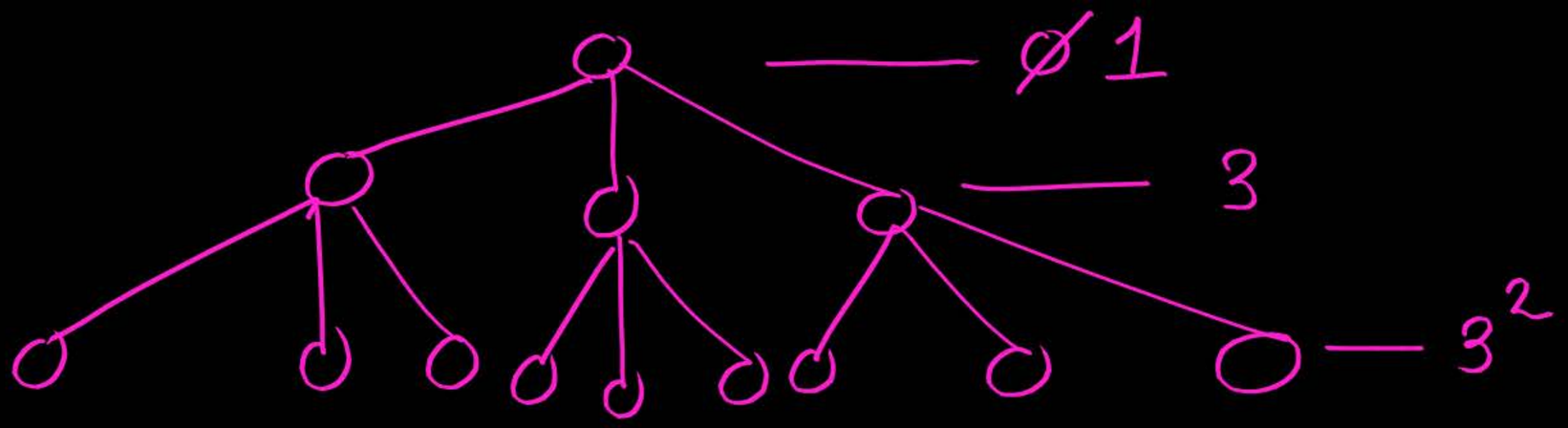
(D) 127 and 7



Slide



$3 \times 6 + 1 = 19$



$$3^0 + 3^1 + 3^2 + 3^3 + \dots + 3^6 = \frac{1(3^7 - 1)}{2} = \frac{2186}{2}$$

$$= 1093$$



Topic : Tree

Theorem
Tree representation
Traversal

Slide

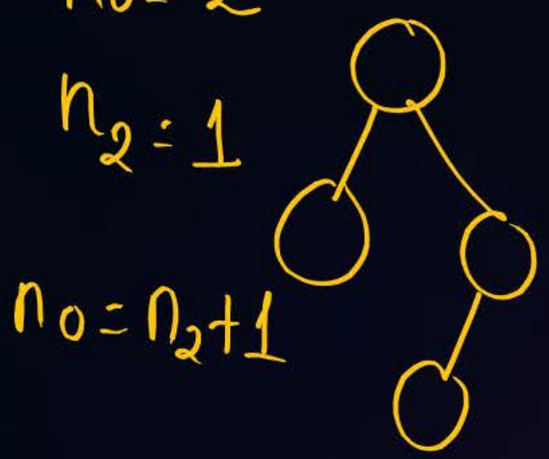


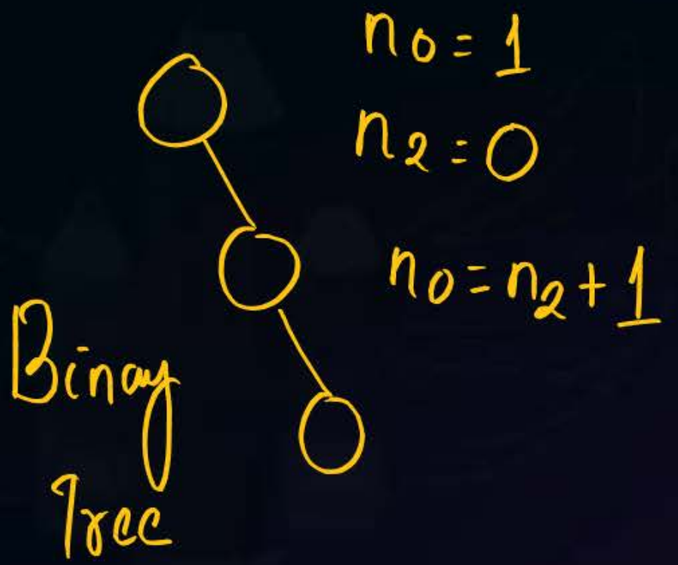
Topic : Tree

Binary Tree : In Binary tree if n_0 is No. of leaf Node

n_1 is No. of Node with 1 children & n_2 is No. of Node with 2 children.

$$n_0 = n_2 + 1$$





Slide



Topic : Tree

degree(n_0) :— 1

degree(n_1) :— 2

degree(n_2) :- 3

Total No. of Nodes :— $n_0 + n_1 + n_2$

Total No. of edges :— $n_0 + n_1 + n_2 - 1$

n_0 - # leaf Node

n_1 - # Node with 1 children

n_2 - # Node with 2 children

Slide



Topic : Tree

Handshaking Algorithm

Sum of degree = $2|E|$

$$n_0 \times 1 + n_1 \times 2 + n_2 \times 3 - 1 = 2(n_0 + n_1 + n_2 - 1)$$

$$\Rightarrow n_0 + 2n_1 + 3n_2 - 1 = 2n_0 + 2n_1 + 2n_2 - 2$$

$$\Rightarrow 2n_0 - n_0 = 3n_2 - 1 - 2n_2 + 2$$

$$\boxed{n_0 = n_2 + 1}$$

Slide



Topic : Tree

$n_0 = n_2 + 1 \Rightarrow 20 = n_2 + 1$

$\rightarrow n_2 = 19$

2015 if in binary tree 20 leaf Nodes are present then No. of Node with 2 children.

Slide



Topic : Tree

Binary Tree Representation

(Heap) : Sequential Representation

An Array is used to store binary tree]

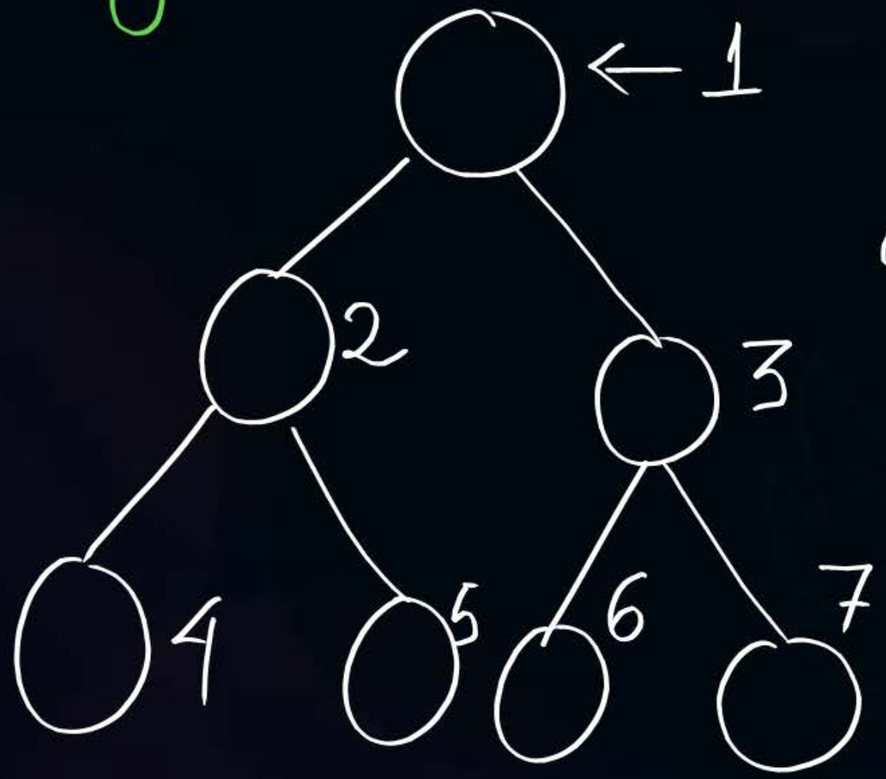
[Index]

Slide



Topic : Tree

In given binary Tree Node position is associated with an Index.



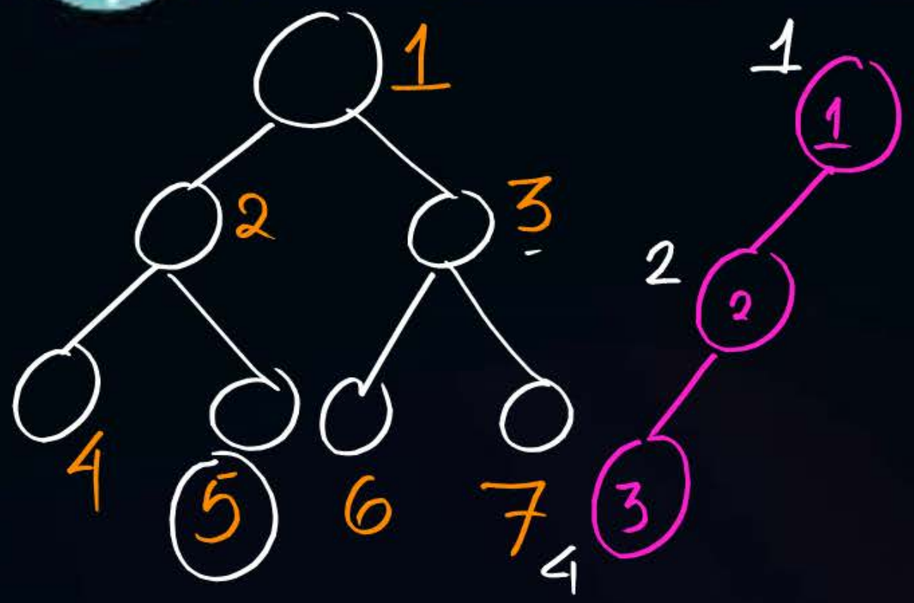
Such that Node data is stored at given Index of array.

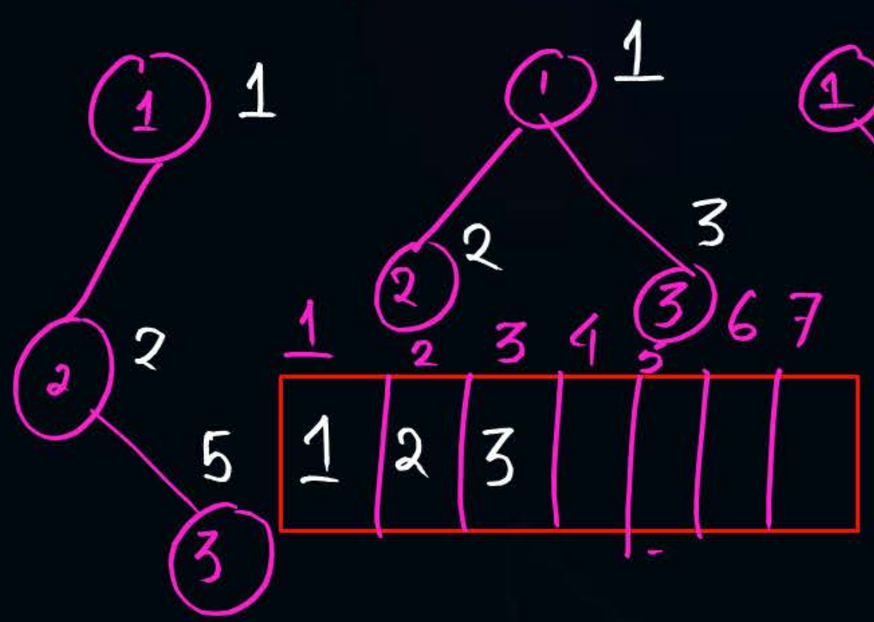
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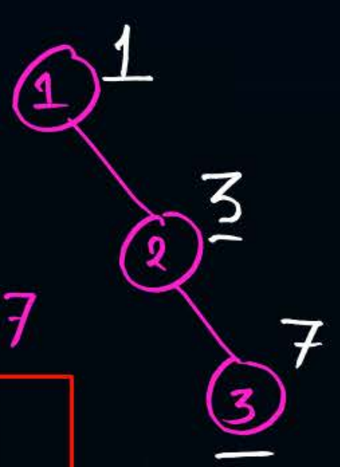


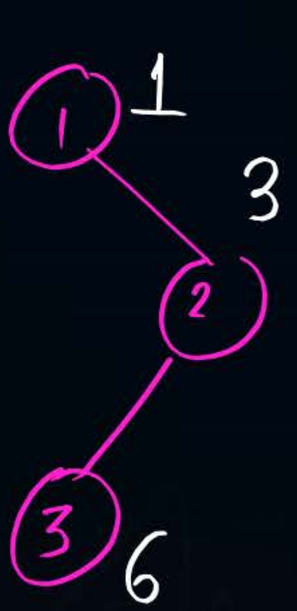
Topic : Tree

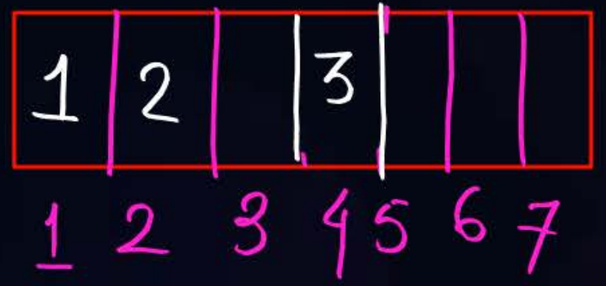
No. of unlabelled tree 3 Node : 5

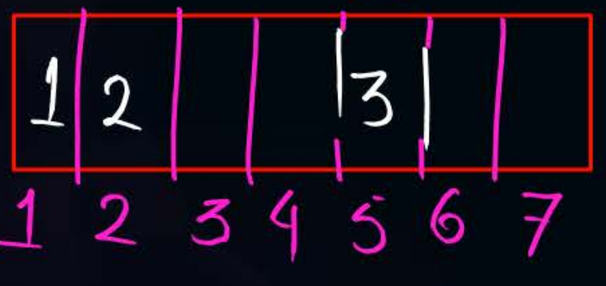




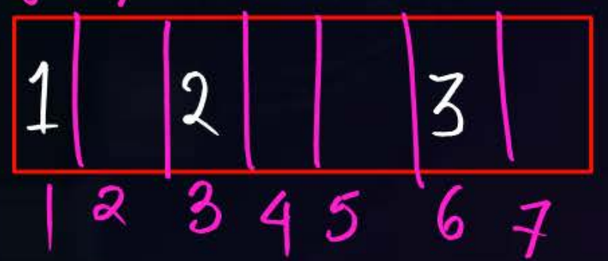












Slide

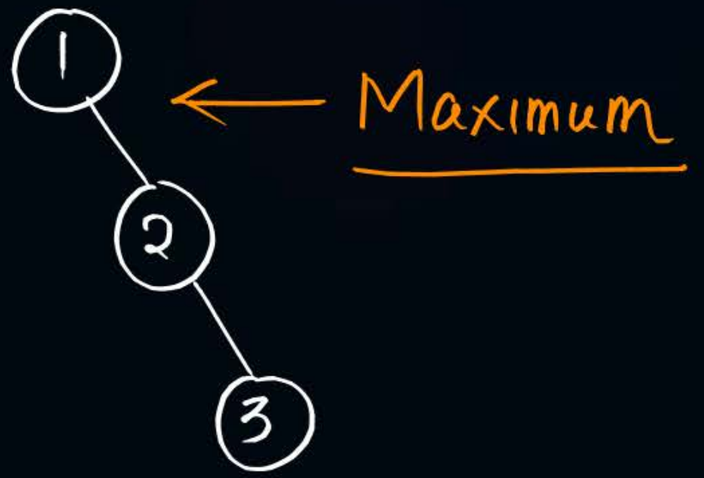


Topic : Tree



Minimum





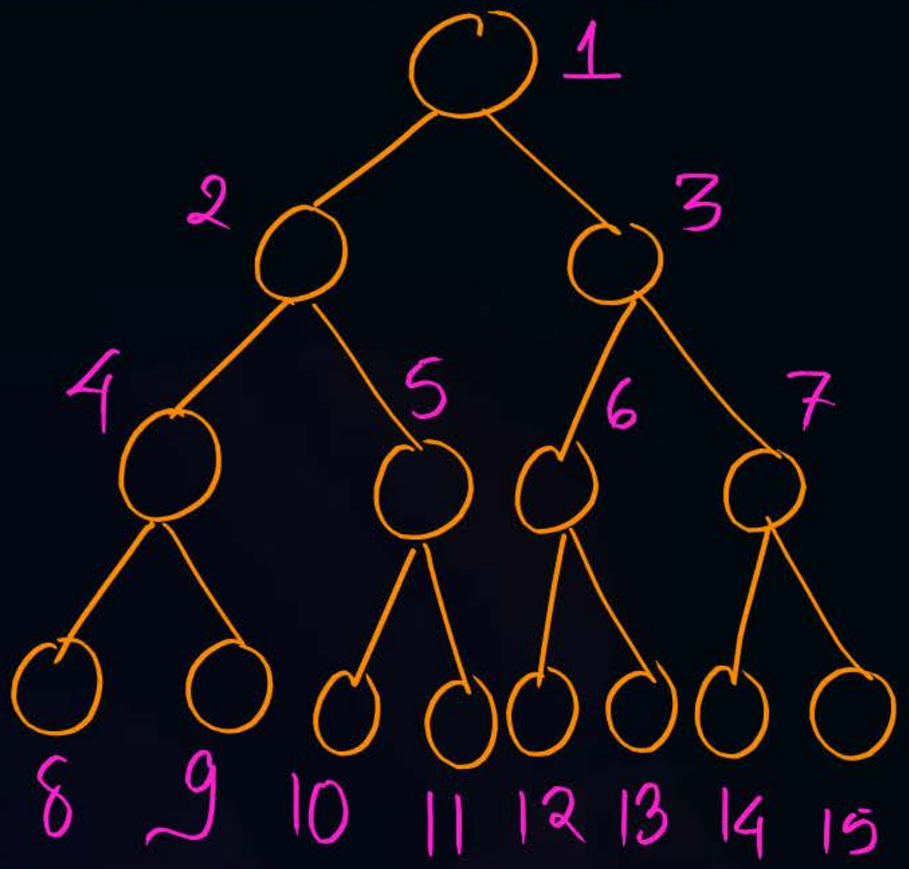
← Maximum



Slide



Topic : Tree



If there is a Node at Index i and $i \neq 1$

then Index of left child : $2i$

Index of right child : $2i+1$

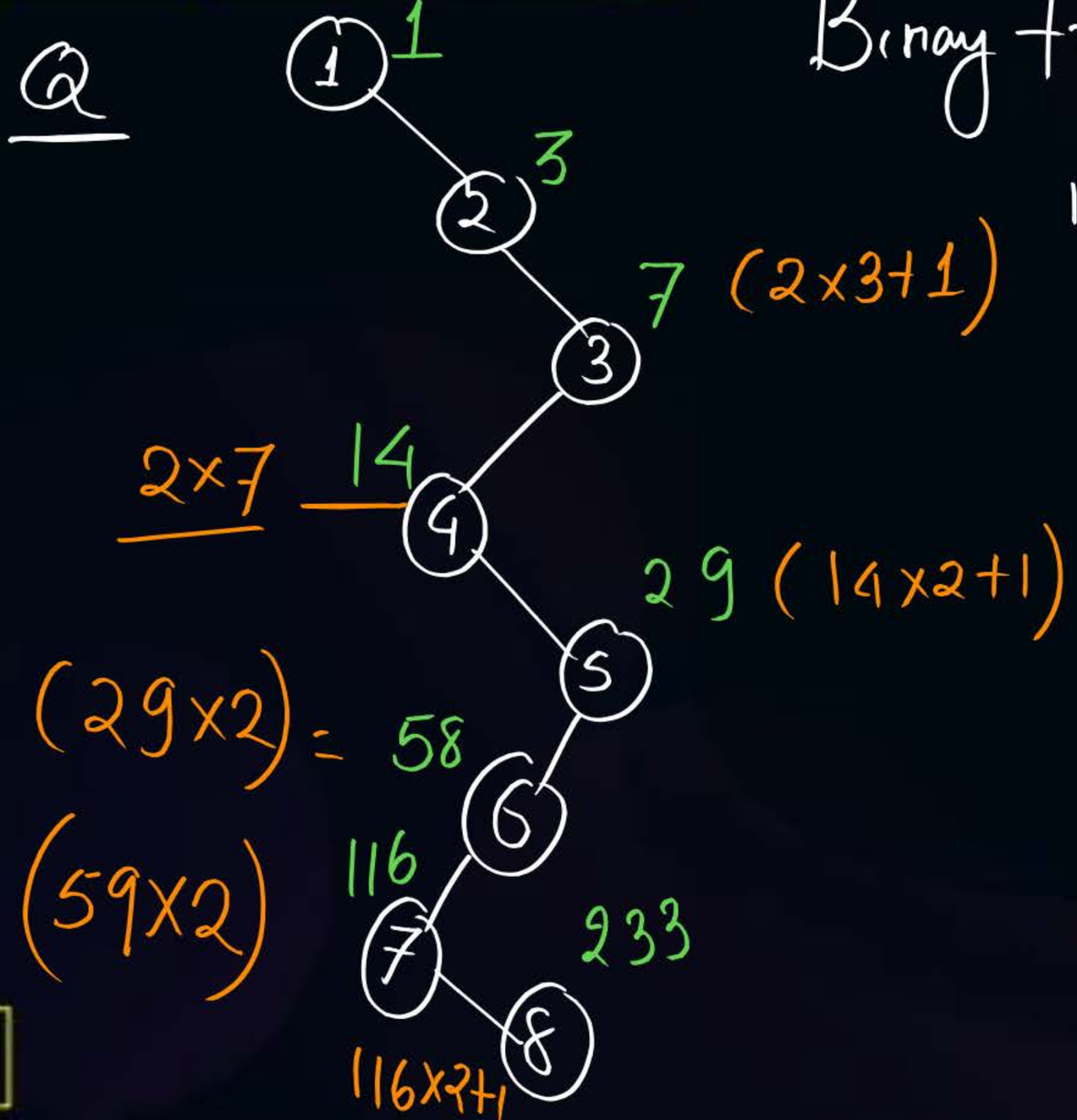
Index of parent : $\lfloor i/2 \rfloor$

Slide



Topic : Tree

#Q



Binary tree

if Index of 8 is x

and Index 6 is y in

Sequential Representation

then x - y is ___

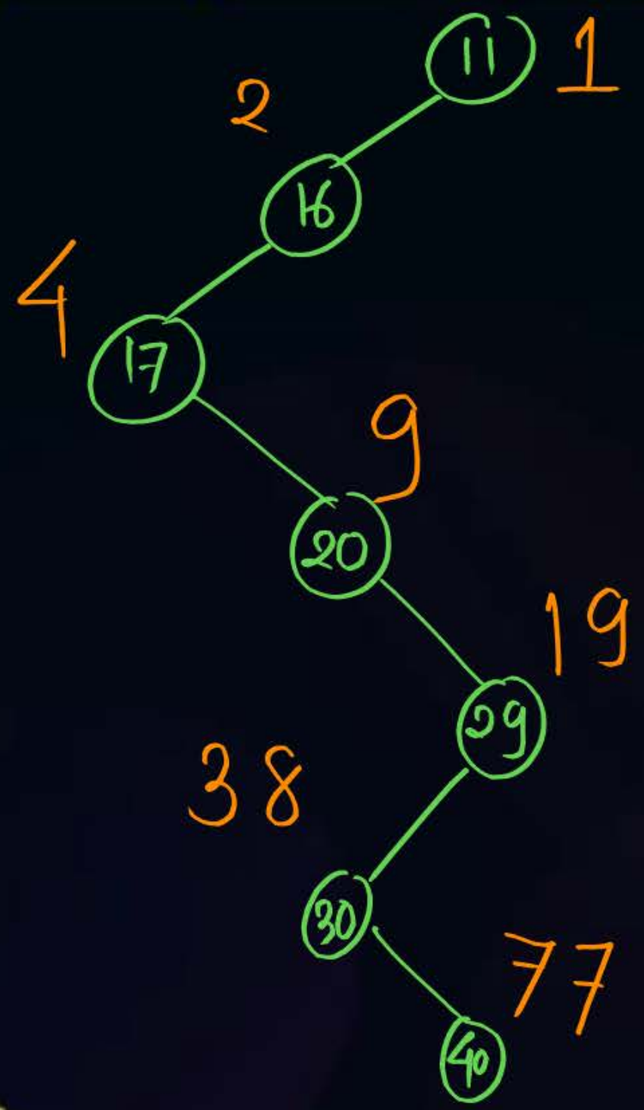
233 - 58

175

Slide



Topic : Tree



if Index start from 1

then if Index of 20 is x - 9

Index of 40 is y the

77 - 9 = 68

y - x is _____

Slide



Topic : Tree

HW

if Index starts from 0 and there is a Node at Index i ($i \neq 0$)

then Index of left child :

Index of Right child :

Index of parent :-

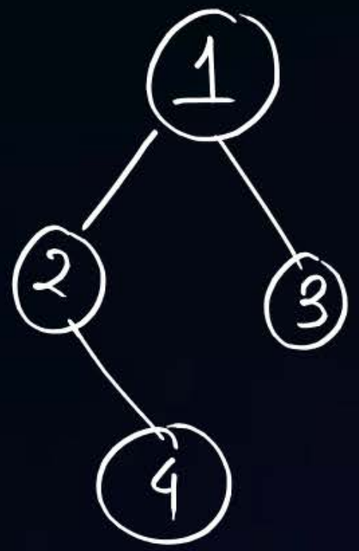
(Heap)

Slide

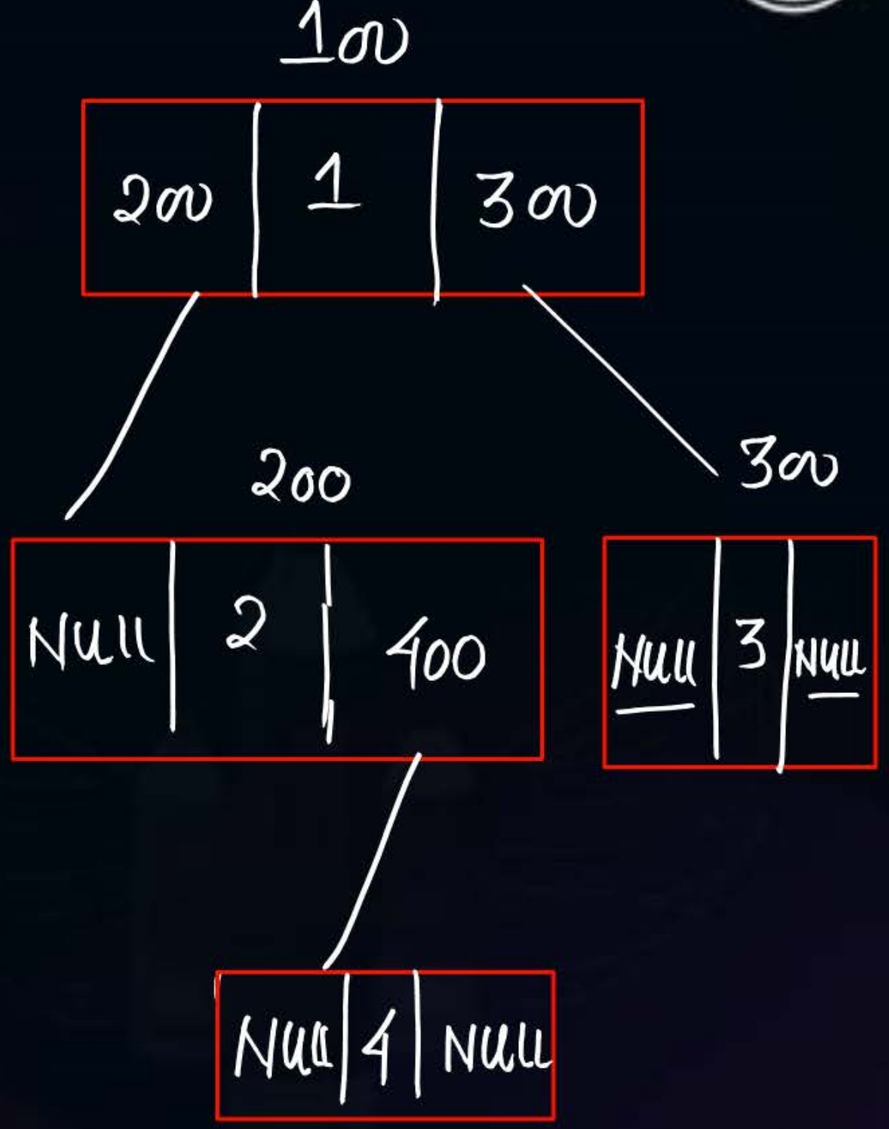


Topic : Tree

Tree Node representation



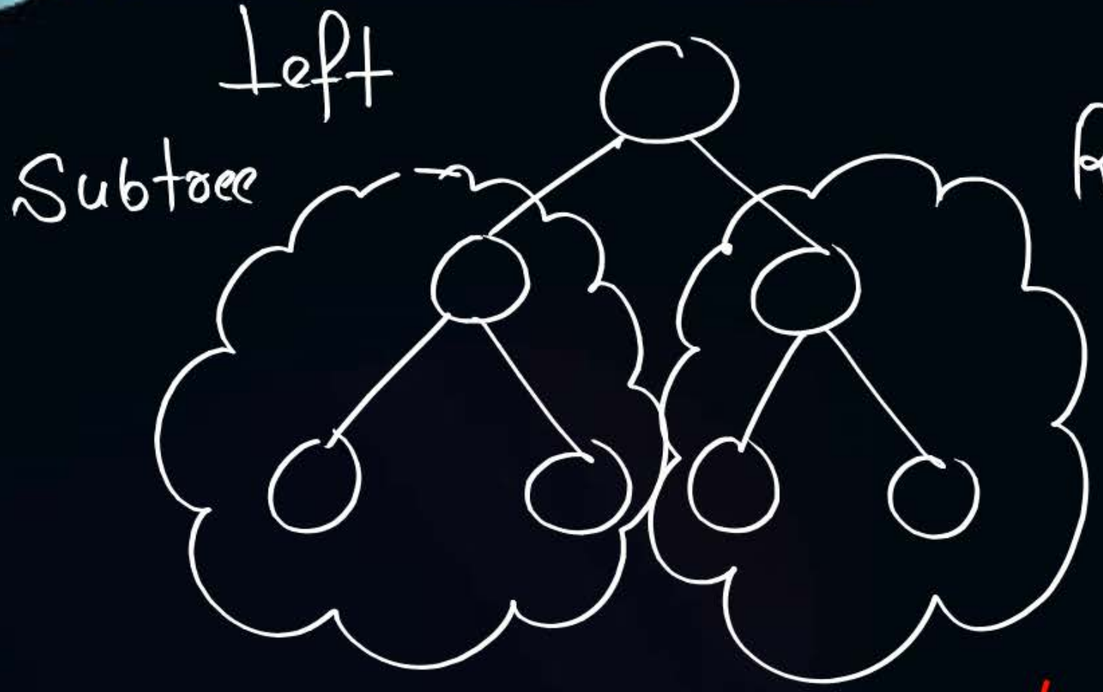

```
typedef struct tnode{
    struct tnode * left;
    int data;
    struct tnode * right;
}Tnode;
```



Slide



Topic : Tree



Tree structure is Recursive

* program written generally Recursive in Nature.

Slide



Topic : Tree

Binary tree Traversal

1. Inorder traversal
2. preorder Traversal
3. postorder traversal

Traversal is a process of visiting every binary tree Node so that we can perform operation on them.

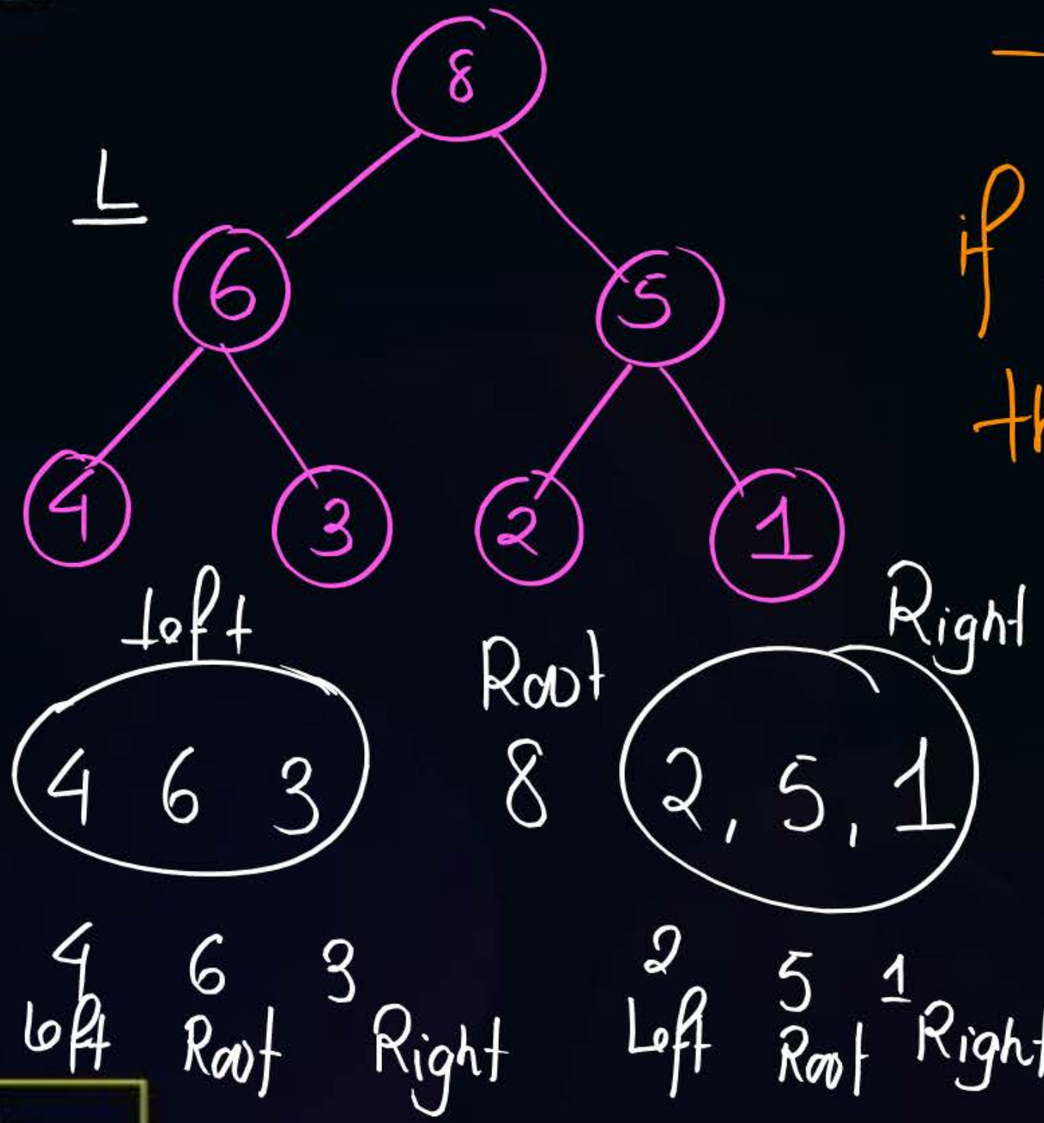
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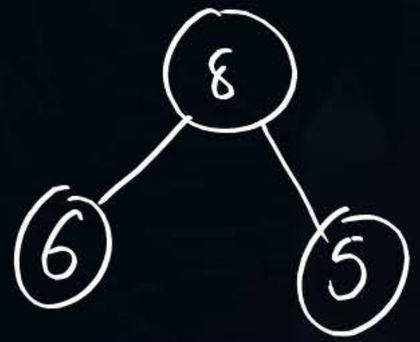


Topic : Tree

Inorder : Left Root Right

if left or right is a Subtree then pattern Left Root Right is repeated.



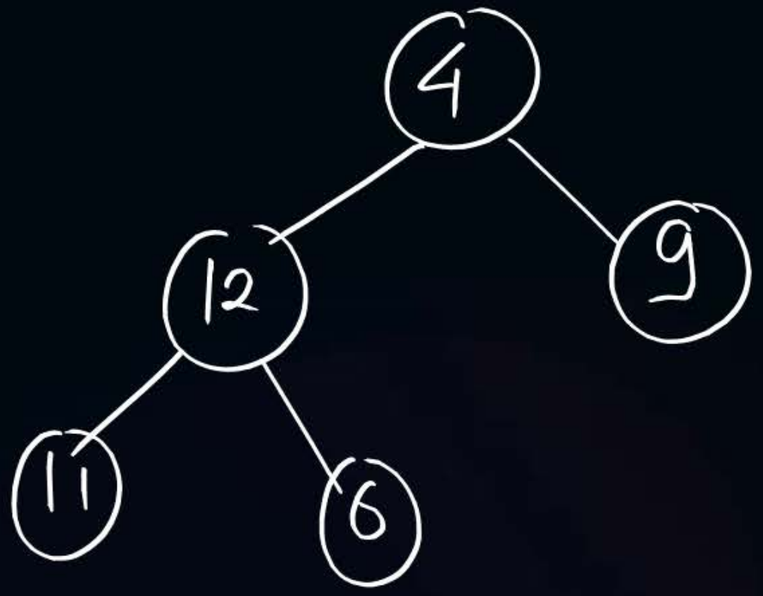


Root
6 8 5
Left Right

Slide



Topic : Tree



Left Root Right

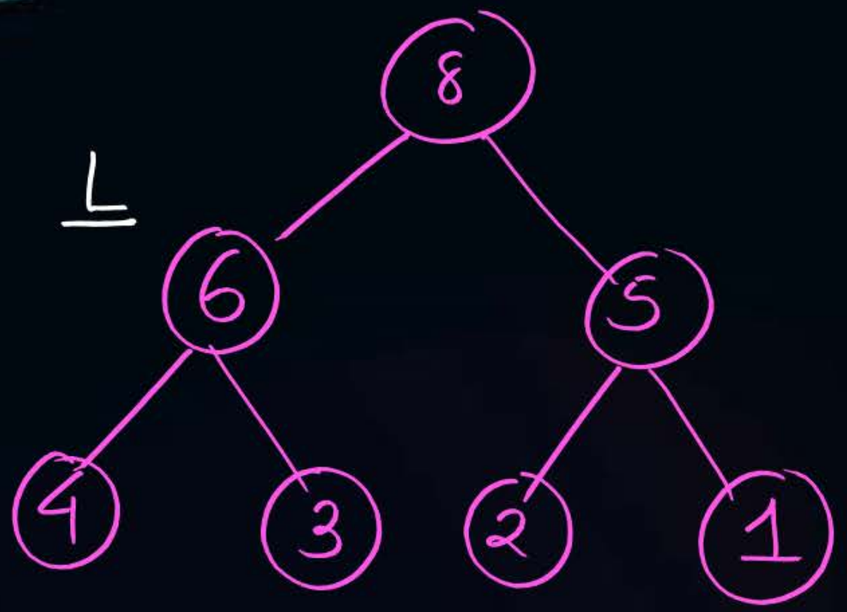
11, 12, 6 4 9

if Node found
the pointer visit
if Subtree found
then
repeat Left Root Right

Slide



Topic : Tree



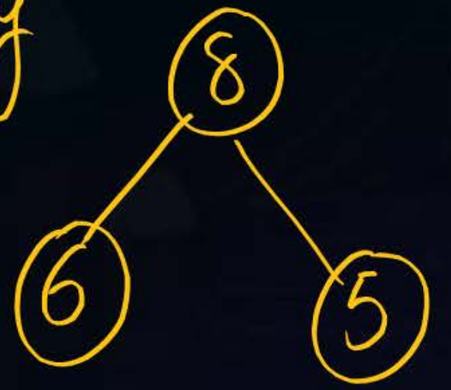
L

preorder : Root Left Right

8 | 6 4 3 | 5 2 1

Root Left Right

Root Identification is easy



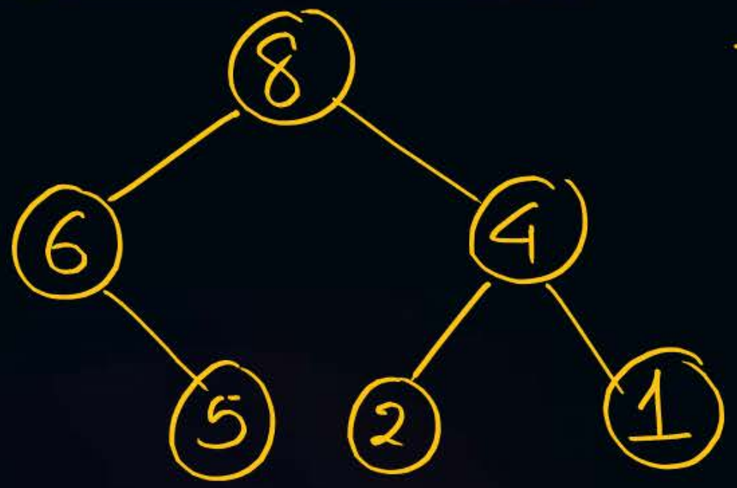
8, 6, 5

Root Left Right

Slide



Topic : Tree



Left Root Right

Inorder :- 6 5 8 2 4 1

preorder :- 8, 6, 5 4, 2, 1

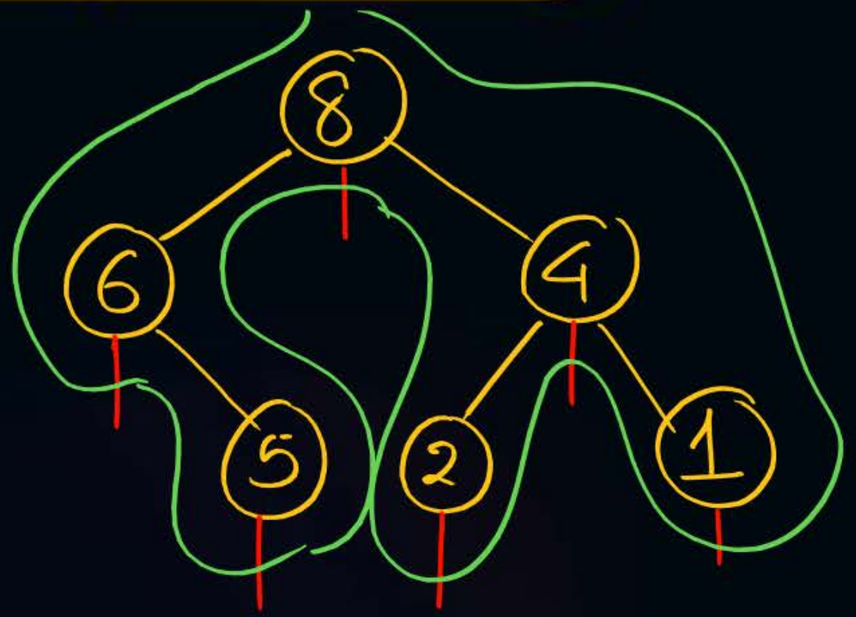
Root left Right

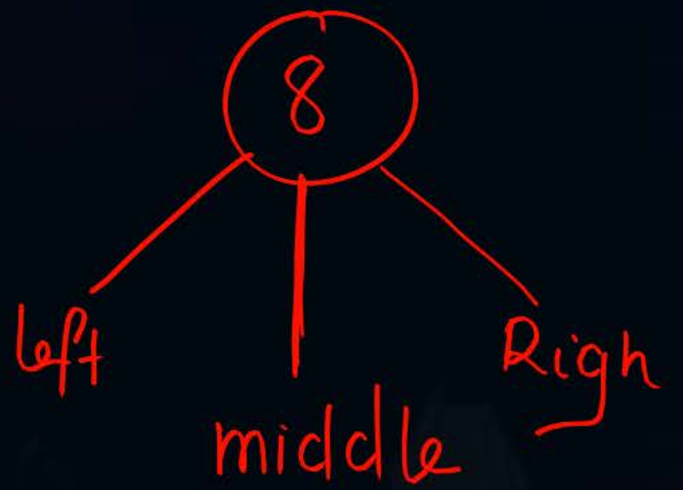
Slide



Topic : Tree

tracing





6, 5, 8, 2, 4, 1 Inorder

Slide

The Tree Node Representation



2 mins Summary

Topic — Theorem

Topic — Tree Representation

Topic — Traversal

Topic

Topic

Slide

THANK - YOU